

HOLOD, Romania

WMO: 151170

Lat: 46.7886N

Lon: 22.1122E

Elev: 540

StdP: 14.41

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.6	(c) 16.8	(d) 8.1	(e) 8.5	(f) 14.3	(g) 13.1	(h) 10.9	(i) 18.6	(j) 23.3	(k) 37.5	(l) 18.5	(m) 36.5	(n) 4.6	(o) 50	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 22.9	(c) 91.8	(d) 71.4	(e) 88.5	(f) 70.9	(g) 85.4	(h) 69.8	(i) 74.1	(j) 86.7	(k) 72.6	(l) 84.4	(m) 71.2	(n) 82.2	(o) 5.3	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.5	114.6	78.8	69.0	108.9	77.1	67.8	104.1	75.4	37.9	86.5	36.5	84.8	35.3	82.2	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.6	(b) 16.2	(c) 13.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 19.6	(b) 16.2	(c) 13.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	31.4	37.6	42.8	52.6	61.6	68.5	70.3	71.5	62.6	52.9	44.9	34.7		
	DBStd	15.59	8.66	8.12	7.56	8.37	6.04	5.78	5.60	5.86	7.62	7.44	8.43	8.22		
	HDD50	1979	578	350	241	72	4	0	0	0	2	58	197	477		
	HDD65	5123	1043	767	687	375	137	31	15	11	136	379	603	939		
	CDD50	2960	0	4	19	149	363	554	631	667	379	147	45	2		
	CDD65	630	0	0	0	3	32	135	181	212	63	3	1	0		
	CDH74	6466	0	0	0	69	381	1272	1793	2240	646	64	1	0		
	CDH80	2330	0	0	0	4	60	397	681	977	209	2	0	0		
(14) Wind	WSAvg	5.3	4.6	5.6	6.7	6.5	5.5	5.1	5.0	4.8	5.0	5.0	5.5	4.6		
(15) Precipitation	PrecAvg	27.9	1.7	1.5	1.7	2.2	2.5	3.4	3.2	2.2	2.6	2.4	2.2	2.4		
	PrecMax	36.9	3.0	4.4	6.5	4.2	5.4	7.4	6.0	4.5	6.9	5.5	4.0	5.6		
	PrecMin	17.0	0.7	0.1	0.3	0.4	0.6	0.6	1.2	0.0	0.3	0.1	0.0	0.0		
	PrecStd	5.6	0.7	0.9	1.4	0.9	1.2	1.5	1.5	1.3	1.6	1.4	0.9	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	60.7	69.8	79.8	84.5	91.3	94.5	96.6	90.6	78.8	70.3	58.5		
		MCWB	49.1	50.2	56.0	61.7	66.5	73.8	71.7	71.8	70.4	62.6	56.9	49.9		
	2%	DB	49.3	56.1	64.0	75.7	81.6	87.9	90.9	92.7	85.2	75.7	66.2	52.4		
		MCWB	45.4	49.3	52.2	60.0	66.8	72.5	71.6	70.9	67.7	61.7	56.4	47.6		
	5%	DB	45.6	52.8	59.0	71.7	78.5	84.9	87.0	89.2	81.7	71.4	61.7	48.4		
		MCWB	43.0	46.8	49.4	58.1	64.3	71.1	70.3	70.3	65.9	60.3	54.2	45.5		
	10%	DB	43.0	49.5	55.0	67.8	75.1	81.3	83.3	85.3	77.1	66.4	57.6	44.8		
		MCWB	41.1	45.3	46.9	55.9	63.5	69.3	68.8	69.6	64.5	58.4	52.5	43.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	52.4	57.6	64.1	69.9	76.4	74.9	75.6	74.3	66.9	59.5	50.9		
		MCDB	54.9	57.4	68.3	75.2	79.3	88.6	88.5	89.3	86.3	73.9	68.1	56.1		
	2%	WB	46.4	50.3	54.2	61.3	68.1	74.0	73.1	73.6	70.1	63.9	57.0	48.5		
		MCDB	48.9	54.3	62.2	73.2	78.0	85.2	85.5	87.3	78.0	70.9	64.3	52.2		
	5%	WB	43.7	48.0	51.0	59.3	66.4	72.2	71.8	72.0	68.0	61.4	55.4	45.6		
		MCDB	45.7	52.2	58.1	69.8	76.1	82.8	82.6	84.9	76.4	68.8	62.1	48.2		
	10%	WB	41.0	45.7	48.5	57.1	64.6	70.5	70.5	70.7	66.0	59.2	52.9	43.2		
		MCDB	42.9	49.2	54.1	66.7	72.8	80.1	80.7	82.8	74.5	65.5	57.6	45.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	14.0	17.9	20.9	20.4	20.6	21.6	22.9	20.2	19.9	15.0	10.0		
		MCDBR	15.4	19.4	23.7	26.7	26.7	25.2	26.7	28.5	27.5	28.6	20.8	14.9		
		MCWBR	11.2	12.7	13.9	13.8	13.0	11.0	9.9	9.7	11.2	15.7	11.8	10.4		
		MCDBR	13.9	17.3	22.1	24.3	22.7	22.6	22.8	24.8	22.3	23.9	18.5	14.0		
	5% WB	MCWBR	10.4	11.5	13.4	12.9	11.8	10.9	9.3	9.1	10.0	14.0	10.8	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.355	0.395	0.448	0.432	0.431	0.448	0.449	0.417	0.381	0.353	0.328		
	taud		2.331	2.279	2.212	2.084	2.195	2.235	2.195	2.189	2.245	2.332	2.358	2.363		
	Ebn at Noon		237	253	260	255	265	266	259	253	250	241	225	222		
	Edh at Noon		24	31	38	48	44	43	44	43	37	29	23	21		
(44) All-Sky Solar Radiation	RadAvg		357	618	1003	1419	1721	1884	1849	1692	1190	795	449	276		
	RadStd		47	89	142	141	165	142	157	157	135	92	54	39		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)	+1.26	N/A	N/A	N/A	+1.80	N/A	N/A	-229	-344	+202	+103

Nomenclature: See separate page